

Vertical, Horizontal and Oblique Asymptotes

Vertical Asymptotes (V. A.)

We can identify Vertical Asymptotes (V. A.) by finding x values that make the denominator ~~undefined~~.

$$= 0$$

If $x = a$ is a V. A., calculate $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$. This limit will either be $+\infty$ or $-\infty$. To decide which test values near the V. A.

Example 1: Sketch the graph of $f(x) = \frac{2(x^2+1)}{(x-1)(x+2)}$ near its vertical asymptotes.

V. A. @ $x = -2$

V. A. @ $x = 1$

$$\lim_{x \rightarrow 1^-} f(x)$$

$$= \lim_{x \rightarrow 1^-} \frac{2(x^2+1)}{(x-1)(x+2)}$$

$$= -\infty$$

$$\lim_{x \rightarrow 1^+} f(x)$$

$$= \lim_{x \rightarrow 1^+} \frac{2(x^2+1)}{(x-1)(x+2)}$$

$$= \infty$$

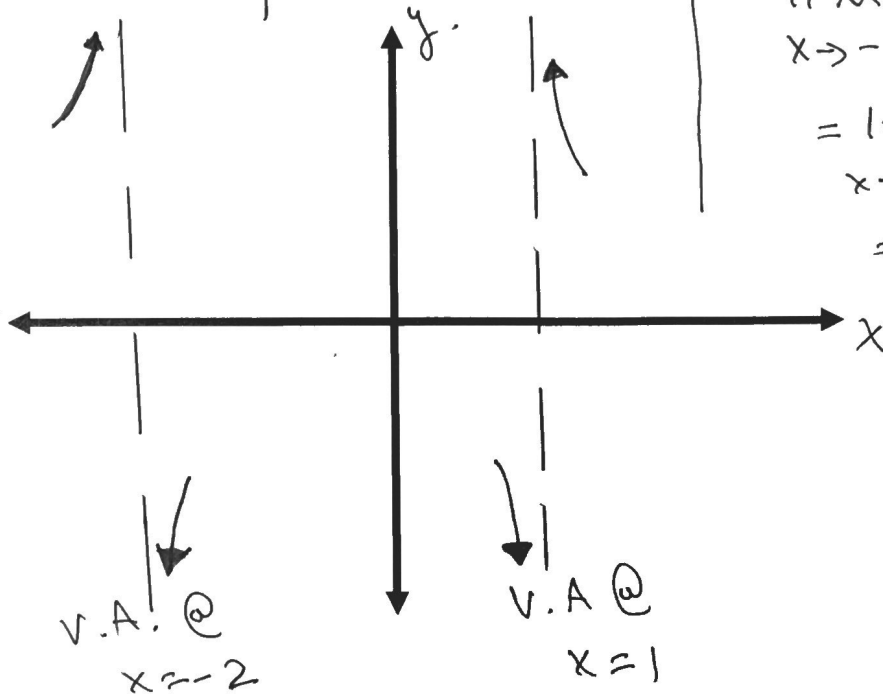
$$\lim_{x \rightarrow -2^-} f(x)$$

$$= \lim_{x \rightarrow -2^-} \frac{2(x^2+1)}{(x-1)(x+2)}$$

$$\lim_{x \rightarrow -2^+} f(x)$$

$$= \lim_{x \rightarrow -2^+} \frac{2(x^2+1)}{(x-1)(x+2)}$$

$$= -\infty$$



Horizontal Asymptotes (H. A.)

- The location of the horizontal asymptotes can be found by determining $\lim_{x \rightarrow \infty} f(x)$ or $\lim_{x \rightarrow -\infty} f(x)$.
- Determine the value of $f(x)$ for both small and large values of x .
 - If the value is greater than the value of the H. A., then the function is approaching the H. A. from above.
 - If the value is less than the value of the H. A., then the function is approaching the H. A. from below.

Example 2: Determine the H. A. of $f(x) = \frac{3-2x}{x-3}$. Also determine where the function is above and below the H. A.

$$\text{H. A. @ } y = -2$$

* Cross test $f(x) = \text{H. A.}$

$$\frac{3-2x}{x-3} = -2$$

$$3-2x = -2x+6$$

$$3 \neq 6$$

$$f(100) = \frac{3-200}{100-3} = \frac{-197}{97} < -2$$

$$f(-100) = \frac{3+200}{-103} = \frac{-203}{103} > -2$$

$\therefore$ As $x \rightarrow \infty$, $f(x) \rightarrow -2^-$ ← Below

As $x \rightarrow -\infty$, $f(x) \rightarrow -2^+$ ← Above

$\Rightarrow$ DOES NOT CROSS H. A.

Example 3: Determine if $f(x) = \frac{3x^2+2x-1}{x^2-4}$ crosses the H. A. If so, determine the point of intersection.

$$\text{H. A. @ } y = 3$$

$$\frac{3x^2+2x-1}{x^2-4} = 3$$

$$3x^2+2x-1 = 3x^2-12$$

$$2x = -11$$

$$x = -\frac{11}{2}$$

$\therefore f(x)$ crosses the H. A.
@ $\left(-\frac{11}{2}, 3\right)$

Example 4: Determine if $f(x) = \frac{x^3 - 13x + 12}{x^2 - 3x - 10}$ crosses the oblique asymptote. If so, determine the point of intersection.

$$\begin{array}{r}
 x^2 - 3x - 10 \overline{) \begin{array}{c} x + 3 \\ x^3 + 0x^2 - 13x + 12 \end{array}} \\
 \underline{-(x^3 - 3x^2 - 10x)} \quad \downarrow \quad \therefore \text{O.A. @} \\
 3x^2 - 3x + 12 \\
 \underline{-(3x^2 - 9x - 30)} \\
 6x + 42 \text{ R.}
 \end{array}$$

$$\begin{aligned}
 R(x) &= 0 \\
 6x + 42 &= 0 \\
 x &= -7
 \end{aligned}$$

$\therefore$ crosses oblique asymptote @ $(-7, -4)$

Example 5: Determine if $f(x) = \frac{2x^3 - x^2 + 3}{x^2}$ crosses the oblique asymptote. If so, determine the point of intersection.

$$\begin{array}{r}
 x^2 \overline{) \begin{array}{c} 2x - 1 \\ 2x^3 - x^2 + 0x + 3 \end{array}} \\
 \underline{-(2x^3)} \quad \downarrow \quad \downarrow \quad \downarrow \quad \therefore \text{O.A. @} \\
 -x^2 + 0x + 3 \\
 \underline{-(-x^2)} \quad \downarrow \quad \downarrow \\
 3 \text{ R.}
 \end{array}$$

$$R(x) = 0$$

$3 \neq 0 \Rightarrow$ Does not cross the oblique asymptote.